

Department of Artificial Intelligence, SVNIT
Subject- Machine Learning (AI301)

LAB ASSIGNMENT-1

Background:

You are given a dataset containing 79 explanatory variables describing (almost) every aspect of residential homes in Ames, Iowa. The target variable is SalePrice. Your task is to perform **data preprocessing** and **exploratory analysis** to prepare the dataset for predictive modeling.

Part A: Data Loading & Basic Inspection

- Load the training dataset into a Pandas DataFrame. Display the first 10 rows.
- How many rows and columns does the dataset have?
- Identify the **numerical** and **categorical** columns. How many are there of each type?
- Check the data types of all columns. Are there any incorrect datatypes that need conversion?

Part B: Univariate Analysis

- For the **numerical columns**:
 1. Plot histograms for at least 5 key features (LotArea, GrLivArea, SalePrice, YearBuilt, OverallQual).
 2. Identify whether the distributions are skewed or approximately normal.
- For the **categorical columns**:
 1. Plot bar charts for at least 3 variables (Neighborhood, HouseStyle, BldgType).
 2. Which categories are most common and which are rare?
- Show value counts for the top 10 most frequent categories across all categorical features.
- Identify any **imbalanced categories** where one category dominates.

Part C: Bivariate Analysis

- Plot a **scatter plot** between GrLivArea and SalePrice. What trend do you observe?
- Plot a **boxplot** of SalePrice across different OverallQual values.
- Create a **correlation heatmap** for the top 10 features most correlated with SalePrice.
- Compare average SalePrice between two neighborhoods of your choice. Is there a significant difference?
- Compare SalePrice between **newer homes** (YearBuilt >= 2000) and **older homes**. Use both:
 1. Boxplot
 2. T-test for mean difference

Part D: Missing Value Treatment

- Find the total number of missing values in each column. List the top 5 columns with the most missing values.
- For **numerical columns**, impute missing values using **median**.
- For **categorical columns**, impute missing values using **mode** or a placeholder like "Missing".
- For `LotFrontage` specifically, impute using the median value grouped by `Neighborhood`.
- After imputation, verify there are no missing values left.

Part E: Outlier Detection & Treatment

- Use the **IQR method** to detect outliers in `LotArea` and `GrLivArea`.
- How many outliers are detected in each feature?
- Remove or cap extreme outliers and explain why you chose that method.
- Replot the scatter plot from Q7 after removing/capping outliers. Has the relationship become clearer?

Part F: Feature Encoding & Scaling

- Encode categorical variables using **One-Hot Encoding** or **Label Encoding**.
- Standardize numerical features using **StandardScaler** or **MinMaxScaler**.
- Save the cleaned and processed dataset as `house_prices_clean.csv`.

Part G: Perform Linear regression

Dataset:

Kaggle – House Prices: Advanced Regression Techniques

